Key Performance Indicators

Customer Support RAG Chatbot

Project Type: Depi Graduation Project

Domain: AI-Powered E-Commerce Customer Support

Team Size: 5 Members

1 KPI Definitions

The following Key Performance Indicators are used to measure the success and effectiveness of the chatbot system. Each KPI is tracked using the project's monitoring stack (Prometheus, MLflow, and Streamlit).

K1 Response Time

PERFORMANCE METRIC

DESCRIPTION	Time taken from receiving a user query to delivering a response
METRIC	`RESPONSE_TIME` Histogram
TRACKED BY	Prometheus
GOAL	Minimize average response latency

K2 Retrieval Confidence

QUALITY METRIC

DESCRIPTION	Relevance score of documents retrieved from the vector database
METRIC	`CONFIDENCE_DISTRIBUTION` Histogram
TRACKED BY	Prometheus, MLflow
GOAL	Maximize % of queries with confidence > 0.5

K3 User Feedback Score

SATISFACTION METRIC

DESCRIPTION	Ratio of positive (thumbs up) to negative (thumbs down) user feedback
METRIC	`FEEDBACK_COUNTER` (positive / negative)
TRACKED BY	Prometheus, Streamlit
GOAL	Maintain positive feedback rate above 80%

K4 Request Throughput

RELIABILITY METRIC

DESCRIPTION	Total number of chat requests handled successfully by the system
METRIC	`REQUEST_COUNTER` (total / by intent)
TRACKED BY	Prometheus, Streamlit
GOAL	Handle concurrent requests with zero downtime

K5 Pipeline Accuracy

AI QUALITY METRIC

DESCRIPTION	Percentage of queries correctly routed through the 5 adaptive pipeline paths (casual, high-confidence, medium-confidence, low-confidence-verified, low-confidence-rejected)
METRIC	Benchmark test suite with 20 test queries across all 5 paths, results logged via `test_adaptive_pipeline.py`
TRACKED BY	MLflow
GOAL	All 5 pipeline paths produce correct routing and appropriate responses

2 Monitoring Stack

The KPIs above are tracked and visualized using a three-layer monitoring architecture integrated into the system.

Prometheus

REAL-TIME METRICS COLLECTION

- Request counters by intent type
- Response time histograms
- Confidence score distribution
- Feedback counters (positive/negative)
- Exposed via `/metrics` endpoint

MLflow

EXPERIMENT TRACKING

- Per-query logging (parameters + metrics)
- Confidence scores per query
- Pipeline path tracking
- Configuration comparison experiments
- Stored locally in `mlruns/`

Streamlit

VISUAL DASHBOARD

- 5 KPI summary cards at top
- Confidence distribution chart
- Response time trend chart
- Feedback breakdown chart
- Recent conversations table

3 KPI Summary

Overview of all defined KPIs, their categories, and tracking tools.

#	KPI	CATEGORY	METRIC	TRACKED BY
K1	Response Time	Performance	`RESPONSE_TIME`	Prometheus
K2	Retrieval Confidence	Quality	`CONFIDENCE_DISTRIBUTION`	Prometheus, MLflow
K3	User Feedback Score	Satisfaction	`FEEDBACK_COUNTER`	Prometheus, Streamlit
K4	Request Throughput	Reliability	`REQUEST_COUNTER`	Prometheus, Streamlit
K5	Pipeline Accuracy	AI Quality	Benchmark suite (5 paths)	MLflow

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